

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.cluster import KMeans
from PIL import Image

data = pd.read_csv(
    'image_396x396_rgb.csv'
)

print("Dataset Shape:")
print(data.shape)

print("\nFirst 5 Pixels:")
print(data.head())

Dataset Shape:
(156816, 5)

First 5 Pixels:
   row  column  R  G  B
0    0      0   0  0  89
1    0      1   0  0  89
2    0      2   1  0  88
3    0      3   1  0  88
4    0      4   2  0  88

pixels = data[
    ['R', 'G', 'B']
].values

image = pixels.reshape(
    396,
    396,
    3
).astype(np.uint8)

print("\nImage Shape:")
print(image.shape)

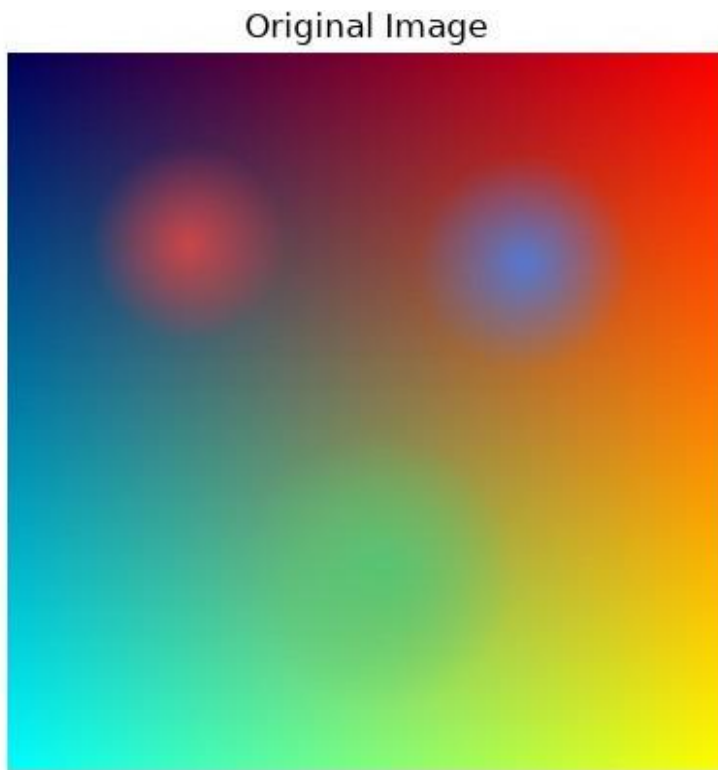
Image Shape:
(396, 396, 3)

plt.imshow(image)

plt.title(
    'Original Image'
)

```

```
plt.axis('off')  
plt.show()
```



```
pixel_data = image.reshape(  
    -1,  
    3  
)  
  
print(  
    "\nPixel Data Shape:",  
    pixel_data.shape  
)  
  
Pixel Data Shape: (156816, 3)  
  
kmeans = KMeans(  
    n_clusters=8,  
    random_state=42,  
    n_init=10  
)  
  
kmeans.fit(  
    pixel_data  
)
```

```
KMeans(n_init=10, random_state=42)

compressed_pixels = (
    kmeans.cluster_centers_[
        kmeans.labels_
    ]
)

compressed_pixels = np.clip(
    compressed_pixels,
    0,
    255
).astype(np.uint8)

compressed_image = compressed_pixels.reshape(
    396,
    396,
    3
)

plt.imshow(
    compressed_image
)

plt.title(
    'K-Means Compressed Image'
)

plt.axis('off')

plt.show()
```

K-Means Compressed Image



```
compressed_image_pil = Image.fromarray(  
    compressed_image  
)  
  
compressed_image_pil.save(  
    'kmeans_compressed_image.png'  
)  
  
print(  
    "\nCompressed image saved successfully!"  
)
```

Compressed image saved successfully!